

THE CERTAINTY STANDARD

Version 1.0 — Specification

A Proposed Evidentiary Framework for Blockchain Attribution Claims

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Status of This Document

Status: Proposed framework. Version 1.0 Specification. This document formalizes the definitions and rules of the Certainty Standard. It has **not** been adopted by any standards body, audit firm, court, or regulator. Adoption is the goal; this document is the foundation.

Empirical status: The reference engine implementing this Specification has been verified against known transaction types (consolidation, CoinJoin, peel chain) via a deterministic validation suite. It has **not** yet been subjected to large-scale adversarial measurement. The central empirical question — whether the Framework-Forced core is non-trivial and decision-relevant under hostile conditions — remains open and is being measured against a pre-registered, frozen pass/kill criterion. Results will be published regardless of outcome.

“No conclusion may be presented without an accounting of the evidentiary assumptions required to reach it.”

1. Purpose and Scope

This Specification defines a disclosure framework for claims made about the control of Bitcoin. Current blockchain attribution providers deliver conclusions — for example, “Entity X controls these funds, confidence 93%” — as a single, opaque score. That score silently blends two categorically different kinds of evidence: facts the protocol verifies, and inferences about how people typically behave.

In cooperative conditions, the blend is serviceable. Under adversarial challenge — cross-examination, a privacy-aware counterparty, or an audit committee demanding support for a balance-sheet figure — the blend conceals exactly what the challenger needs to know: which part of the conclusion is provable, and which part is a guess. The purpose of this Specification is to require that the two be separated and disclosed.

The scope of this Specification is the classification and disclosure of attribution claims. It is not an investigative methodology, does not rank the accuracy of competing heuristics, and does not determine legal ownership.

2. Control, Not Ownership

This framework certifies claims about **control** — the cryptographic capacity to spend — and never about **ownership**, which is a legal conclusion.

Control is a protocol-level concept: possession of the keys required to produce a valid spend.

Ownership is a legal concept assigned by courts, contracts, and statutes. The two routinely diverge: a trustee controls assets it does not own; a custodian controls customer assets; a thief controls stolen assets; an exchange controls deposits belonging to thousands of users.

No output of this framework constitutes a determination of legal ownership. The framework supplies the control evidence upon which qualified professionals — auditors, counsel, courts — may base ownership determinations within their own authority and duty of care.

3. The Evidence Separation Principle

Every attribution claim must be decomposed into four declared components:

Component	Definition
Forced Facts	Claims entailed under one of the two forcing grades defined in Section 4.
Inferred Findings	Probabilistic outputs of behavioral heuristics, declared as such, with the heuristic identified.
Contradictory Evidence	Signals that weaken or defeat the proposed conclusion, graded by severity (Section 5).
Residual Uncertainty	The declared delta between the Forced Facts and the conclusion actually being offered.

A conclusion presented without this decomposition is, under this framework, an undisclosed opinion — not evidence.

4. Two Grades of Forcing (Formal Definitions)

This framework distinguishes two strictly different grades of forced claims. Conflating them would quietly broaden the meaning of “forced” after defining it narrowly; they are named separately to foreclose that.

Definition 4.1 — Protocol-Forced. A claim is Protocol-Forced if and only if its negation would render the observed transaction invalid under Bitcoin consensus rules — that is, the claim is strictly necessary for the transaction to exist on the chain at all. **The boundary is validity, not accuracy.** Any claim whose negation is consistent with a valid transaction — however improbable — is not Protocol-Forced, regardless of how reliable it has historically been.

Definition 4.2 — Framework-Forced. A claim is Framework-Forced if it is affirmatively established by at least one Tier-1 heuristic of a declared, versioned heuristic set; survives challenge from the full conflict-set of that heuristic set; and is reached without counting silence as agreement. Framework-Forced claims are conditional on the set and its version. They are the strongest claims an engine can certify, and they are explicitly weaker than Protocol-Forced claims.

Every classification must state which grade it asserts. A Framework-Forced claim presented as Protocol-Forced is non-compliant on its face.

Requirement 4.3 — Adversarial composition. A declared heuristic set must contain heuristics capable of contradicting one another. A set composed of mutually agreeable heuristics produces a falsely large forced core and is non-compliant.

5. Graduated Contradiction

Contradictory evidence is not uniform. The framework recognizes three grades:

5.1 Weak contradiction — reduces confidence in an inferred finding; must be disclosed; does not change classification.

5.2 Material contradiction — places a finding in open dispute; the claim may not be presented as a finding, only as a contested signal.

5.3 Claim-failing contradiction — defeats a forced classification entirely; the claim reverts to inferred or undetermined status.

6. The Three Tenets

6.1 Forced vs. Inferred. A hard, declared boundary separates forced claims from behavioral inference.

6.2 Silence is not agreement. The absence of contradictory signals is an epistemic gap, never affirmative proof. All gaps must be declared.

6.3 Survival, not assertion. A claim is classified as forced only if it is affirmatively established and survives challenge from the full declared conflict-set.

7. The Standard and the Engine

This framework deliberately separates two entities. **The Standard** governs disclosure: what must be declared, decomposed, and graded before a conclusion may be presented. It is implementation-agnostic and can be satisfied by any forensics provider, including incumbents. **The Engine** is any specific implementation — a particular heuristic set, its tiers, its constants, and its conflict rules — that produces classifications compliant with the Standard.

Critiques of any individual heuristic are critiques of an Engine, not of the Standard. Engines are versioned and replaceable; the disclosure obligation is permanent.

8. Classification Outcomes

A compliant Engine reports each decision-relevant claim as exactly one of the following:

Forced-Control Positive (FC+). The protocol entails that a valid spend was authorized by the controlling key(s) for the inputs in question, and the linkage of those inputs survives the declared conflict-set. The protocol proves the spend-event; it does not, by itself, identify which person or entity held the key at the moment of signing. Mapping a key to a real-world party is always an additional claim requiring its own declared evidence (custody records, signed messages, account attestations). An FC+ certificate is a claim about keys and transactions — never, on its own, about persons.

Forced Non-Linkage (FN-). Non-linkage is entailed under the declared heuristic set and version, and survives the conflict-set. FN- certificates are explicitly conditional: they certify that no compliant heuristic in the declared set can establish linkage — not that linkage is impossible under all future methods. Every FN- certificate must state the set and version against which it is valid.

Contested (FLIP). Compliant heuristics disagree; no forced classification may be issued.

Undetermined. All heuristics are silent; the silence is declared, not resolved.

Reported yield must state FC+ and FN- composition **separately**. They are different products with different buyers and must never be aggregated into a single figure.

9. Framework Governance

A standard whose heuristic selection is discretionary has merely relocated hidden judgment from confidence scores into set composition. To foreclose that critique, the following governance rules are part of the Standard:

- 9.1 Versioning.** Every heuristic set is versioned. No classification exists except relative to a named, numbered set.
- 9.2 Change logging.** Additions, removals, and modifications of heuristics are logged with date, rationale, and author. The log is part of the Standard’s record.
- 9.3 Reproducibility.** Prior classifications remain reproducible under the set-version that produced them. New versions supersede old ones visibly; they do not silently overwrite history.
- 9.4 Mandatory conflict.** A proposed set is non-compliant unless it contains heuristics capable of contradicting one another.
- 9.5 Citation requirement.** Every report, certificate, and audit output must identify the exact Engine and set-version used. An undated, unversioned classification is non-compliant on its face.
- 9.6 Grade declaration.** Every certified claim states its forcing grade (Protocol-Forced or Framework-Forced). Grade inflation is a compliance violation, not a stylistic choice.

Without governance, standards become politics. With governance, standards become infrastructure.

10. Intended Use

This framework is designed for audit review, litigation support, regulatory evaluation, expert testimony, and forensic quality assurance. It is not intended to replace investigative methodologies; it is intended to disclose their evidentiary composition. A forensics provider, an auditor relying on a provider’s report, and a litigator challenging one can all apply the framework to the same conclusion — that symmetry is the point.

11. Limitations and Disclaimers

This framework documents evidence so that professionals may discharge their own duty of care. It does not absorb, transfer, or insure anyone’s liability; it does not determine legal ownership; and it is not an attestation service within the meaning of regulated public-accounting practice.

All forced claims are forced relative to a declared heuristic set, never absolutely. A specific known limitation: certain privacy techniques (for example, undetectable Payjoin constructions) are designed to be indistinguishable from ordinary payments and therefore impose a permanent epistemic ceiling on heuristic-based linkage claims. The framework accommodates this by requiring that such limits be declared rather than hidden.

This Specification has not been empirically validated at scale. A framework that cannot survive its own disclosure requirements does not deserve adoption.

12. Defined Terms (Quick Reference)

Term	Meaning
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Protocol-Forced	Claim whose negation would make the transaction invalid under consensus rules.
Framework-Forced	Claim established by a Tier-1 heuristic and surviving the declared conflict-set.
FC+	Forced-Control Positive: control entailed and surviving challenge.
FN-	Forced Non-Linkage: non-linkage entailed under the declared set and version.
FLIP	Contested: compliant heuristics disagree; no forced classification issued.
Undetermined	All heuristics silent; silence declared, not resolved.
Conflict-set	The set of declared heuristics capable of contradicting a claim.
Engine	A specific, versioned implementation producing Standard-compliant classifications.